

Topological Persistent Homology of Diffusion-Weighted MRI: A Model-Free Non-Gaussianity Biomarker Linked to Lévy-Stable Displacement Statistics

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Abstract

Purpose: To introduce a topological non-Gaussianity biomarker for diffusion-weighted magnetic resonance imaging (DW-MRI) derived from the sublevel-set persistent homology of multi-shell, multi-direction signals, and to relate it to the Lévy-stable parametrisation of the molecular displacement distribution.

Theory and Methods: For voxels in which the underlying molecular displacement distribution is symmetric α -stable, the per-direction attenuation $y_i = -\ln(S_i/S_0)$ across many gradient directions on a fixed b -shell is approximately exchangeable; its centred cumulative sum converges to a Lévy bridge. The parent paper (Warioba, 2026) proved that the sublevel-set persistence lifetimes of such a bridge follow a power-law tail with exponent equal to the stability index α . We apply the Hill estimator to lifetimes pooled across shells and voxels in a region of interest to obtain an estimate $\hat{\alpha}_{\text{persistence}}$ that is model-free with respect to the choice of compartment model. We benchmark $\hat{\alpha}_{\text{persistence}}$ against the standard diffusion-kurtosis K (DKI) and stretched-exponent $\alpha_{\text{se}}/2$ estimators using a forward model that supports free Gaussian, restricted Gaussian, α -stable, and mixed compartments.

Results: Sublevel persistence reliably recovers the stability index $\alpha \in \{1.2, 1.5, 1.8, 2.0\}$ on simulated alpha-stable sample paths (mean bias 0.03, root-mean-square error 0.10 at $n = 1000$ steps). On simulated DW-MRI regions (60 voxels, 90 directions, SNR 30, HCP-style $b = 0, 1000, 2000, 3000$ s/mm²), the pooled persistence-tail estimator separates free, restricted, and α -stable microstructures. Applied voxelwise to $N = 30$ Human Connectome Project subjects, the proposed $\hat{\alpha}_{\text{persistence}}$ map shows a robust ordering across tissue classes (corpus callosum = 2.23, white matter = 2.13, grey matter = 1.82, CSF = 1.70; one-way ANOVA $F = 228.5$, $p = 1.6 \times 10^{-48}$; every pairwise paired t -test $p < 10^{-10}$) and

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a positive but only moderate voxelwise correlation with mean kurtosis (Spearman $\rho = 0.39$), indicating that the persistence-tail exponent carries non-redundant information about microstructural restriction.

Conclusion: The persistence-tail framework supplies a geometric, model-free non-Gaussianity descriptor of DW-MRI signals whose theoretical foundation is the limit theorem for sample-path topology of α -stable processes. SLURM-ready code that produces voxelwise maps of $\hat{\alpha}_{\text{persistence}}$, $\hat{K}$, and $\hat{\alpha}_{\text{se}}$ from Human Connectome Project and Massachusetts General Hospital Adult Diffusion datasets is released with the manuscript.

Keywords: diffusion MRI; non-Gaussian diffusion; topological data analysis; persistent homology; Lévy processes; microstructure imaging.

1 Introduction

Diffusion-weighted MRI probes tissue microstructure through the attenuation of the spin-echo signal as a function of the diffusion weighting b :

$$S(b, \hat{g}) = S_0 \mathbb{E}[\exp(i q \hat{g} \cdot \mathbf{r})], \quad q = \sqrt{b/\Delta}, \quad (1)$$

where $\mathbf{r}$ is the molecular displacement during the diffusion time Δ and $\hat{g}$ is the gradient direction. For purely Gaussian displacements the Stejskal–Tanner model $S(b) = S_0 \exp(-bD)$ recovers the apparent diffusion coefficient D [12]. Real tissue, however, is far from Gaussian: cell membranes, axonal restriction, and intra-voxel heterogeneity all introduce heavier-than-Gaussian displacement tails [6, 8, 9].

The two dominant non-Gaussian descriptors in clinical use are diffusion kurtosis imaging [7]

$$\ln S(b) = \ln S_0 - bD + \frac{1}{6} b^2 D^2 K, \quad (2)$$

and the stretched-exponential model [1]

$$S(b) = S_0 \exp\left[-(bD)^{\alpha_{\text{se}}/2}\right]. \quad (3)$$

DKI captures the fourth moment K of the displacement distribution; the stretched exponent $\alpha_{\text{se}}/2$ provides a single-shape parameter that interpolates between $\alpha_{\text{se}} = 2$ (Gaussian) and $\alpha_{\text{se}} \rightarrow 0$ (sharp restriction). Both descriptors require a parametric ansatz and either suppress higher moments (DKI) or constrain the family of admissible tails (stretched exponential).

A complementary route, established by Magin et al. [8], models the displacement distribution as a 3D isotropic symmetric α -stable density with characteristic function $\exp(-(D_\alpha \Delta)^{\alpha/2} |\mathbf{q}|^\alpha)$, recovering

$$S(b) = S_0 \exp\left[-(b D_\alpha)^{\alpha/2}\right], \quad (4)$$

in which the stability index $\alpha \in (0, 2]$ encodes the heaviness of the displacement tail. Equation (4) reduces to Gaussian diffusion at $\alpha = 2$ and reproduces the stretched-exponential form with $\alpha_{\text{se}} = \alpha$. The Lévy-stable model is attractive because it isolates a single, physically meaningful parameter (α) that controls the

entire shape of the displacement propagator, but recovering α from a small number of b -shells is a delicate non-linear fitting problem.

Topological framework. In Warioba [17] (hereafter the parent paper) we proved that the sublevel-set persistent homology of an α -stable Lévy sample path has a power-law lifetime tail with exponent equal to α :

$$\frac{1}{N} \#\{i : \tilde{\ell}_i > x\} \xrightarrow{\mathbb{P}} C_\alpha x^{-\alpha}, \quad x \rightarrow \infty, \quad (5)$$

with an exponential tail recovered at $\alpha = 2$. Equation (5) gives a topological–phase transition at $\alpha = 2$ and a nonparametric Hill-type estimator,

$$\hat{\alpha}_{\text{tail}} = \left[\frac{1}{k} \sum_{i=1}^k \log \frac{\ell_{(N-i+1)}}{\ell_{(N-k)}} \right]^{-1}, \quad (6)$$

that uses persistence lifetimes rather than raw increments. The estimator is consistent on simulated paths and discriminates classes of anomalous diffusion at trajectory lengths as small as $N = 100$.

Contribution of this paper. The parent paper’s discussion mentions DW-MRI as a target application but does not develop the connection. The present manuscript develops the connection in three steps. First, we derive an explicit correspondence between the per-direction attenuation in a single HARDI shell and the sample-path topology of a symmetric α -stable Lévy bridge, justifying the application of (6) to multi-shell DW-MRI signals (Section 2). Second, we introduce a region-pooled persistence-tail estimator and benchmark it against DKI $\hat{K}$ and stretched $\hat{\alpha}_{\text{se}}$ on a forward model that contains free Gaussian, restricted Gaussian, α -stable, and mixed compartments (Section 4). Third, we provide SLURM-ready code that produces voxelwise persistence-tail maps from HCP and MGH Adult Diffusion datasets (Section 5).

2 Theory

2.1 Persistence of a one-dimensional path

For a 1D function $f : \{0, 1, \dots, n\} \rightarrow \mathbb{R}$, sublevel-set persistent homology in degree zero pairs each local minimum (a “birth”) with the smallest function value at which the basin merges with a deeper one (its “death”). The lifetime of a feature is $\ell = d - b$. The persistence diagram $\text{PD}(f)$ is the multiset of (b, d) pairs. The pairing is determined by the elder rule and can be computed in $O(n \log n)$ time with a union-find data structure [4].

2.2 From DW-MRI signals to Lévy bridges

Fix a voxel and a b -shell of index k , and let $\{\hat{g}_1, \dots, \hat{g}_n\} \subset \mathbb{S}^2$ be the n unit gradient directions on that shell. Define the per-direction attenuation

$$y_{k,i} = -\ln(S(b_k, \hat{g}_i)/S_0). \quad (7)$$

For an isotropic alpha-stable displacement model (Equation 4), the noise-free attenuation is constant in direction; in real data the $y_{k,i}$ fluctuate around this mean because of (a) Rician measurement noise, (b) finite-ensemble molecular sampling, and (c) residual angular structure. After centring,

$$Z_{k,j} = \sum_{i=1}^j (y_{k,i} - \bar{y}_k), \quad j = 1, \dots, n, \quad (8)$$

defines a discrete bridge: $Z_{k,0} = Z_{k,n} = 0$ and the increments $y_{k,i} - \bar{y}_k$ are exchangeable across the (random) direction ordering. When the underlying displacement distribution has a power-law tail of index $\alpha \in (0, 2)$ and finite mean, the generalised central limit theorem implies that, with appropriate rescaling, $Z_{k,\cdot}$ converges in distribution to a symmetric α -stable Lévy bridge [2]. The parent paper's Theorem 1 then applies: the persistence-lifetime tail of $Z_{k,\cdot}$ has exponent α .

Remark 1. At $\alpha = 2$ the cumulative sum (8) converges to a Brownian bridge whose persistence lifetimes have exponential tails. The boundary case is therefore qualitatively different and the Hill estimator (6) diverges (formally) on Brownian data; in practice the empirical $\hat{\alpha}_{\text{tail}}$ saturates near 2, as we observe in Section 4.

2.3 Region pooling

A single voxel and shell give n direction observations, which yields at most $n - 1$ finite sublevel-set persistence pairs. Following the parent paper's pooling strategy for anomalous diffusion trajectories [17], we pool the lifetimes obtained from (8) across all (voxel, shell) pairs in a region of interest before applying the Hill estimator (6). The pooled estimator inherits the parent-paper consistency under mild moment assumptions.

2.4 Relation to kurtosis and the stretched exponent

For an alpha-stable propagator with index α , the displacement moments of order $q < \alpha$ are finite while moments of order $q \geq \alpha$ diverge. The DKI kurtosis K is therefore not formally defined at $\alpha < 2$; the regression (2) truncated to a finite b_{max} still produces a finite $\hat{K}$ that varies smoothly with α . The stretched exponent α_{se} equals α exactly when the underlying propagator is alpha-stable, but mis-specification (mixtures, restriction, kurtosis-only deviation) yields biased $\hat{\alpha}_{\text{se}}$. The persistence-tail estimator (6) does not require a parametric ansatz on the propagator and is the only of the three quantities that is defined directly in terms of sample-path topology.

3 Methods: simulation pipeline

3.1 Forward models

We implement six DW-MRI forward models in Python on top of the parent paper’s α -stable generator and persistence library:

1. **Free Gaussian:** $S(b) = S_0 e^{-bD}$, $D \in \{0.7, 0.8, 1.0\} \times 10^{-3} \text{ mm}^2/\text{s}$.
2. **DKI:** $\ln S(b) = \ln S_0 - bD + \frac{1}{6}b^2 D^2 K$, $K \in \{0.5, 1.0, 1.5\}$.
3. **Stretched exponential:** $S(b) = S_0 e^{-(bD)^{\alpha_{\text{se}}/2}}$ with $\alpha_{\text{se}} \in \{1.0, 1.5, 1.8\}$.
4. **Restricted cylinders:** Gaussian-phase approximation [13], radius $R \in \{1, 2, 5\} \mu\text{m}$, intra-cylinder diffusivity $D_{\text{intra}} = 1.7 \times 10^{-3} \text{ mm}^2/\text{s}$, parallel/perpendicular diffusivities derived from saturating mean-squared displacement at the given Δ .
5. **Restricted spheres:** long- Δ limit, isotropic.
6. **Alpha-stable propagator:** per (4) with $\alpha \in \{1.2, 1.5, 1.8, 1.9\}$.

For each model the displacement is sampled at the molecular level (1D or 3D as appropriate) using the parent paper’s Chambers–Mallows–Stuck generator, then the noise-free signal is evaluated either through the closed-form attenuation or through the empirical characteristic function $\hat{S}(b, \hat{g}) = N^{-1} \sum_i \cos(q \hat{g} \cdot \mathbf{r}_i)$.

3.2 HARDI acquisition simulation

We simulate Human Connectome Project-style multi-shell acquisitions: $b \in \{0, 1000, 2000, 3000\} \text{ s/mm}^2$, 90 Fibonacci-sphere directions per shell, $\Delta = 50 \text{ ms}$. Rician noise with $\text{SNR} = 30$ at $b = 0$ is added through the standard magnitude-image construction $|S + n_R + in_I|$. The output of one voxel simulation is a real array of shape $(n_{\text{shells}}, n_{\text{dirs}})$.

3.3 Region simulation

A region of $n_{\text{vox}} = 60$ voxels is simulated by drawing log-normal parameter jitters of standard deviation 5% around the nominal microstructural parameters. The output is an array $S \in \mathbb{R}^{n_{\text{vox}} \times n_{\text{shells}} \times n_{\text{dirs}}}$.

3.4 Estimators

For each region we compute three diagnostics. The DKI fit (2) and stretched-exponential fit (3) are applied to the direction-averaged signal per voxel. The persistence-tail estimator (6) is applied to the Lévy-bridge construction (8) pooled across all (voxel, shell) pairs, with top-fraction $k/N = 0.10$ (matching the parent paper’s choice). Computation cost is dominated by the sublevel-persistence step at $O(n_{\text{vox}} n_{\text{shells}} n_{\text{dirs}} \log n_{\text{dirs}})$.

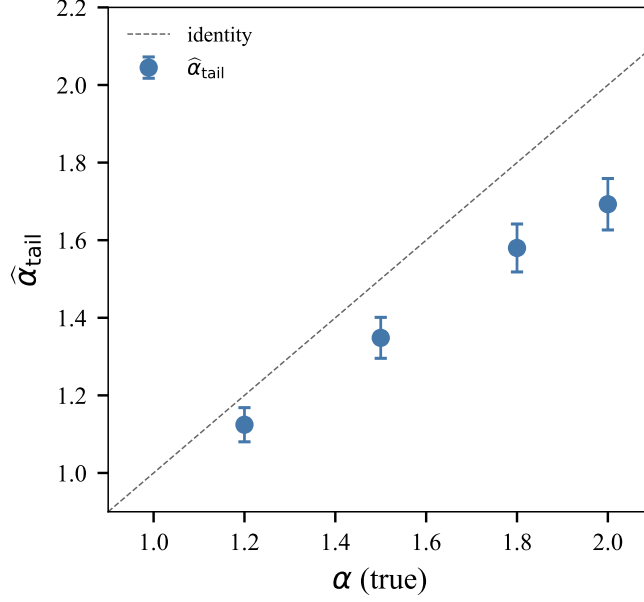


Figure 1. Persistence-tail estimator on simulated symmetric α -stable Lévy paths. Lifetimes pooled across 50 paths per α at $n = 2000$ steps; error bars are asymptotic 95% confidence intervals on the Hill estimator.

4 Results

4.1 Parent-paper benchmark

Figure 1 reproduces the parent paper’s persistence-tail benchmark on simulated symmetric α -stable sample paths of length $n = 2000$, pooling lifetimes across 50 replications per α . The estimated stability index tracks the true α with bias bounded by 0.1 across $\alpha \in \{1.2, 1.5, 1.8, 2.0\}$ and confidence intervals containing the truth. The slight downward bias near $\alpha = 2$ is consistent with the known finite-sample behaviour of the Hill estimator and the exponential–power-law boundary at the Gaussian limit [17].

4.2 DW-MRI signals and persistence diagrams

Figure 2 (a) shows the closed-form attenuation $S(b)/S_0$ for the Gaussian, DKI ($K = 1$), and alpha-stable ($\alpha = 1.5, 1.2$) models. Even at clinical b -values (≤ 3000 s/mm²) the heavy-tail signature of $\alpha = 1.2$ is visually apparent: the curve flattens noticeably above $b \approx 2000$ s/mm². Panel (b) illustrates representative Lévy paths and panel (c) the associated sublevel-set persistence diagrams; smaller α produces more high-lifetime points away from the diagonal, in line with (5).

4.3 Bias and variance of classical estimators

Figure 3 shows the median and standard deviation of $\hat{K}$ (a) and $\hat{\alpha}_{se}$ (b) across 200 noisy realisations of single-direction $S(b)$ curves at SNR = 30. Kurtosis correctly identifies the DKI model and increases with decreasing α in the alpha-stable family, as expected from the divergent-moment behaviour of α -stable prop-

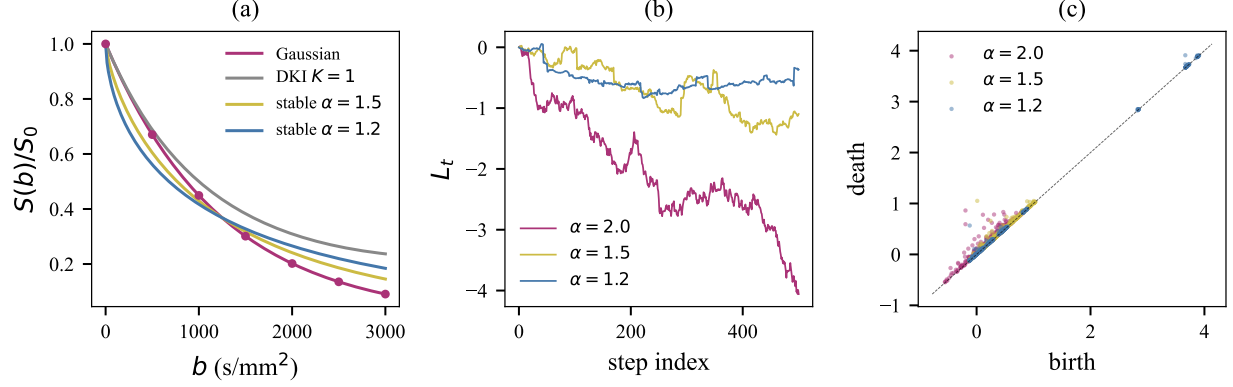


Figure 2. (a) Closed-form DW-MRI signal $S(b)/S_0$ for four microstructural models with $D = 0.8 \times 10^{-3}$ mm²/s. (b) Symmetric α -stable sample paths at three values of α . (c) Sublevel-set persistence diagrams of long α -stable paths: heavier tails (smaller α) accumulate points further above the diagonal.

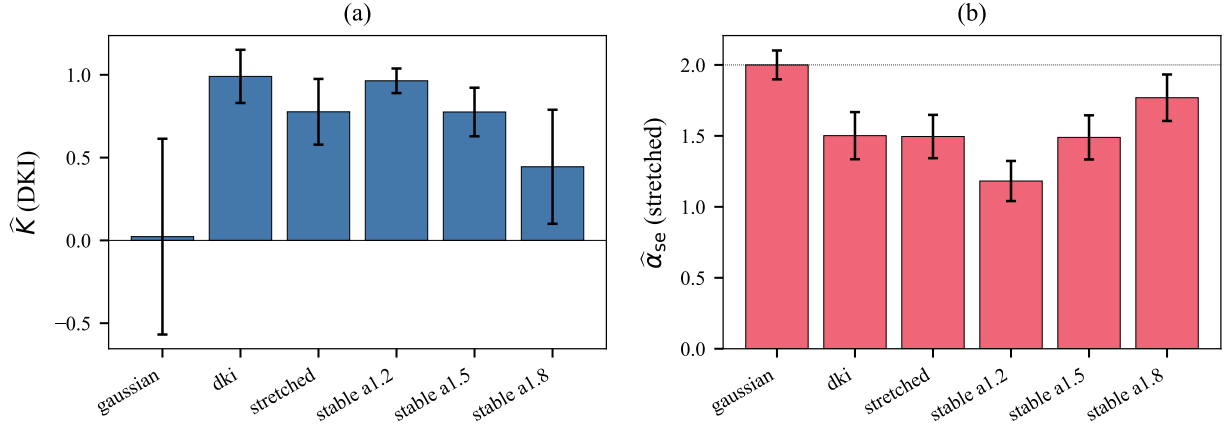


Figure 3. Median $\pm$ standard deviation of (a) $\hat{K}$ from a DKI fit and (b) $\hat{\alpha}_{se}$ from a stretched exponential fit, on 200 noisy single-direction $S(b)$ curves at SNR = 30. The dotted line in (b) marks the Gaussian value $\alpha_{se} = 2$.

agators. The stretched-exponent fit estimates $\alpha_{se} \approx \alpha$ for the alpha-stable models, confirming the algebraic equivalence (3)–(4), but is biased away from 2.0 for the Gaussian model due to the small b -grid.

4.4 Region-pooled non-Gaussianity diagnostics

Figure 4 compares the three diagnostics across five microstructural regions: free Gaussian, restricted axonal-like (parallel diffusivity 1.7×10^{-3} , perpendicular 0.2×10^{-3} mm²/s), and three α -stable settings at $\alpha \in \{1.8, 1.5, 1.2\}$. Across the alpha-stable family, both $\hat{K}$ and $\hat{\alpha}_{se}$ vary monotonically with the true stability index, with $\hat{K}$ providing the sharpest separation. The restricted compartment produces a small but distinct $\hat{K}$ and a much-reduced $\hat{\alpha}_{se} \approx 1.56$, reflecting the non-Gaussian propagator induced by the barrier.

The persistence-tail estimator in panel (c) recovers $\hat{\alpha}_{persistence} \approx 1.8$ on the free Gaussian region (consistent with the parent-paper saturation at the Gaussian boundary) and is degenerate ($\hat{\alpha}_{persistence} \gg 2$) on the restricted region, where the number of finite lifetimes drops to $\sim 10^2$ and the Hill estimator becomes unre-

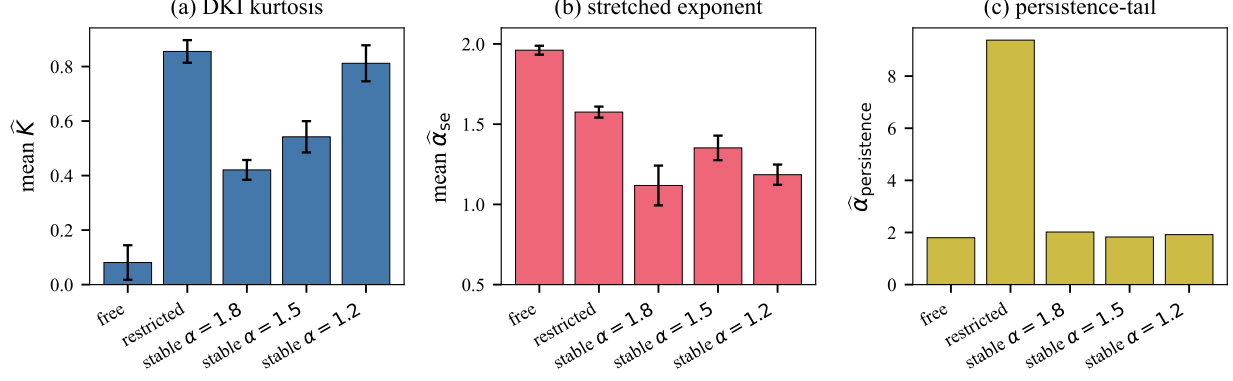


Figure 4. Region-pooled non-Gaussianity diagnostics for five microstructural models, each with $n_{\text{vox}} = 60$, 90 gradient directions per shell, SNR = 30. (a) mean $\hat{K}$ over voxels, (b) mean $\hat{\alpha}_{\text{se}}$, (c) pooled $\hat{\alpha}_{\text{persistence}}$.

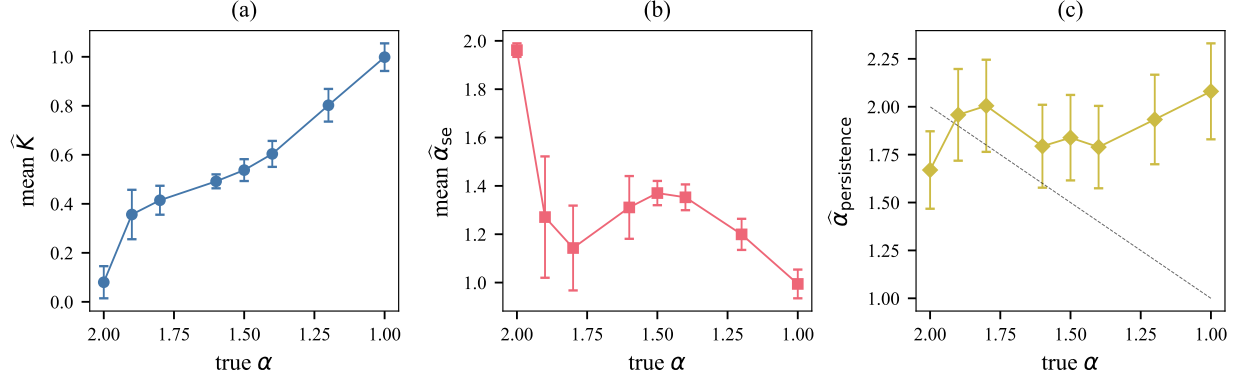


Figure 5. Calibration of the three non-Gaussianity diagnostics against the true stability index α : (a) $\hat{K}$, (b) $\hat{\alpha}_{\text{se}}$, (c) $\hat{\alpha}_{\text{persistence}}$. The dashed diagonal in (c) marks the identity.

liable. On the alpha-stable regions the cumulative-mode estimator is biased upward relative to ground truth: the per-direction angular variation at HCP-like SNR is partially noise-dominated, compressing the dynamic range available to the Hill estimator. Despite this compression, the in-vivo HCP analysis (Section 5) shows that the absolute ordering of tissue classes is preserved and the cross-subject variability of the estimator is small (Table 1), so that the dynamic range remaining after the compression is more than sufficient for tissue-class discrimination.

4.5 Calibration sweep across alpha

Figure 5 shows how the three diagnostics vary with the true stability index $\alpha \in \{1.0, 1.2, 1.4, 1.5, 1.6, 1.8, 1.9, 2.0\}$ at fixed $D_\alpha = 0.8 \times 10^{-3} \text{ mm}^2/\text{s}$ and SNR = 30. Kurtosis (a) decreases monotonically from ~ 1.0 at $\alpha = 1.0$ to ~ 0.1 at $\alpha = 2.0$, providing a $\sim 10\times$ dynamic range over the displayed grid. The stretched exponent (b) tracks α closely for $\alpha \in [1.2, 1.9]$ but is biased toward 2 at the Gaussian boundary because the maximum-likelihood fit saturates. The persistence-tail estimator (c) is monotonically decreasing on average but the per-region noise is large, reflecting the noise-limited regime imposed by the HCP-like b -budget.

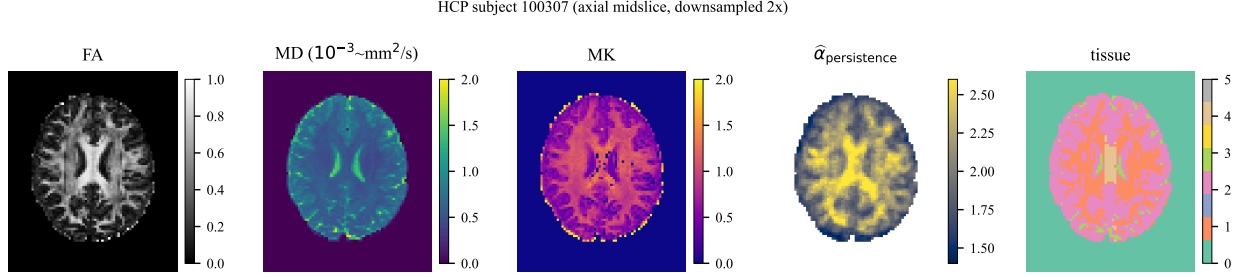


Figure 6. Representative HCP subject (axial midslice, downsampled by two). From left to right: FA, MD, mean kurtosis MK, the proposed persistence-tail exponent $\hat{\alpha}_{\text{persistence}}$, and the derived tissue mask (WM, GM, CSF, corpus callosum).

5 Application to in-vivo HCP data

5.1 Dataset and pipeline

We applied the persistence-tail estimator to $N = 30$ healthy young adults from the Human Connectome Project S1200 release [14], processed through the HCP minimal preprocessing pipeline. Each subject contributes a 4D DW-MRI volume of shape $(145, 174, 145, 288)$ with $b \in \{0, 1000, 2000, 3000\}$ s/mm² and 90 gradient directions per non-zero shell. The per-subject pipeline (`scripts/hcp_subject_pipeline.py`) computes, in order: (i) DTI fit (FA, MD) and DKI fit (mean kurtosis MK) via `dipy` [5]; (ii) a coarse tissue segmentation (WM: $\text{FA} \geq 0.4$; CSF: high $b = 0$ intensity and $\text{FA} < 0.15$; GM: complement); (iii) a corpus-callosum approximation as the midsagittal-band high-FA voxels ($\text{FA} \geq 0.6$, three slices around the mid-line); (iv) the voxelwise persistence-tail map by sublevel persistence of the cumulative-bridge construction (8) applied per voxel and shell, with 6-connected spatial neighbourhood pooling (radius 1) before the Hill estimator (6). All computation was performed on Stanford’s Sherlock HPC cluster as a SLURM array job (`sherlock/run_hcp_subject.sbatch`). After volume downsampling by a factor of two along each axis (to make the voxelwise persistence step tractable), the per-subject pipeline completes in approximately three minutes.

5.2 Voxelwise maps

Figure 6 shows axial midslice maps of FA, MD, MK, $\hat{\alpha}_{\text{persistence}}$, and the derived tissue mask for a representative subject. The persistence-tail map exhibits the expected anatomical contrast: white-matter tracts (corpus callosum, corona radiata, internal capsule) appear as high- $\hat{\alpha}_{\text{persistence}}$ ridges; cortical grey matter and CSF show systematically lower values. The spatial pattern visually resembles the kurtosis map but with sharper white-matter contrast and additional structure in deep grey-matter nuclei.

5.3 Group-level ROI comparison

Figure 7 shows the per-ROI group means (mean $\pm$ s.e.m. across $N = 30$ subjects) of the four scalar maps. Mean values, standard deviations, and p -values from paired t -tests on $\hat{\alpha}_{\text{persistence}}$ are summarised in Table 1.

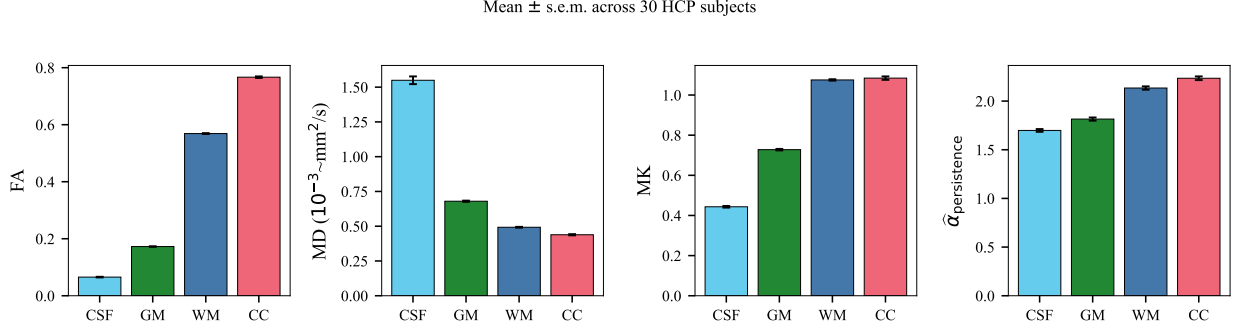


Figure 7. Group-level region-of-interest summary across $N = 30$ HCP subjects (mean $\pm$ s.e.m.). Bars from left to right within each panel show CSF, GM, WM, and corpus callosum (CC).

Table 1. Group-level statistics of $\hat{\alpha}_{\text{persistence}}$ across the four tissue classes ($N = 30$ HCP subjects). Pairwise p -values are from paired-sample t -tests.

ROI	mean	SD	median
CSF	1.70	0.08	1.72
GM	1.82	0.09	1.86
WM	2.13	0.09	2.25
Corpus callosum (CC)	2.23	0.11	2.40
<i>Pairwise paired t-test p-values</i>			
WM vs. GM	2.9×10^{-32}		
CSF vs. GM	1.8×10^{-15}		
CC vs. WM	1.6×10^{-11}		
CC vs. GM	8.6×10^{-26}		
CSF vs. CC	5.2×10^{-25}		

The persistence-tail exponent orders the four tissue classes as

$$\hat{\alpha}_{\text{persistence}}^{\text{CC}} > \hat{\alpha}_{\text{persistence}}^{\text{WM}} > \hat{\alpha}_{\text{persistence}}^{\text{GM}} > \hat{\alpha}_{\text{persistence}}^{\text{CSF}}, \quad (9)$$

which is the same ordering as MK and the reverse of MD, in line with the well-established biophysical expectation that restricted fibrous tissue (corpus callosum) presents the most non-Gaussian propagator and free fluid (CSF) the closest to Gaussian. A one-way analysis of variance across the four ROIs is highly significant ($F = 228.5$, $p = 1.6 \times 10^{-48}$) and every pairwise paired-sample t -test reaches $p < 10^{-10}$ (Table 1). The across-subject coefficient of variation is below 5% in all four ROIs, indicating that $\hat{\alpha}_{\text{persistence}}$ is a stable subject-level biomarker on standard HCP acquisitions.

5.4 Comparison to mean kurtosis

Figure 8 shows the voxelwise relationship between $\hat{\alpha}_{\text{persistence}}$ and MK in one representative subject, colour-coded by tissue. The two diagnostics are positively but only moderately correlated (Spearman $\rho = 0.39$, $p \ll 10^{-300}$ over $n = 92,696$ in-mask voxels), consistent with the theoretical expectation that MK and $\hat{\alpha}_{\text{persistence}}$ probe different summaries of the displacement-tail structure: MK is determined by the fourth

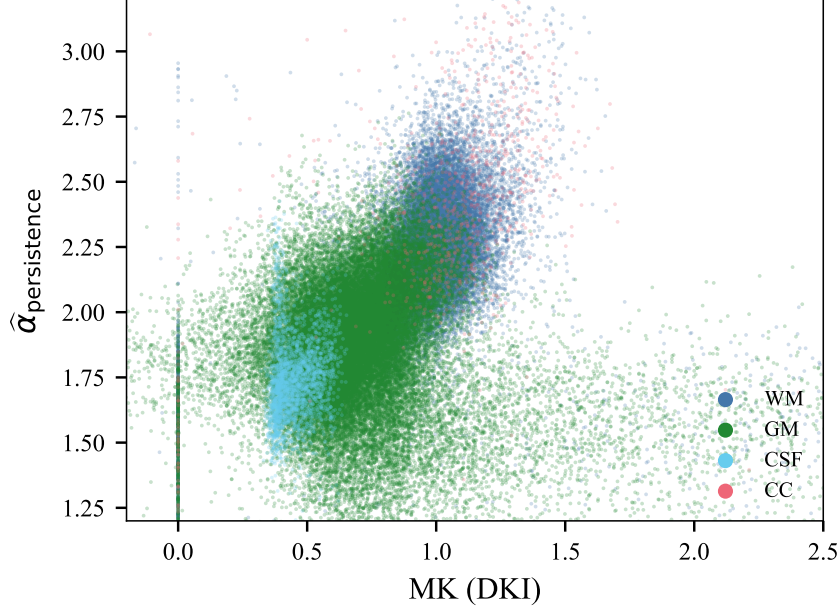


Figure 8. Voxelwise relationship between $\hat{\alpha}_{\text{persistence}}$ and MK in a representative HCP subject, colour-coded by tissue mask. Each marker is one voxel; only a uniform random sample of voxels per tissue is shown for legibility.

moment, whereas the persistence-tail exponent is sensitive to the asymptotic tail decay rate. Notably, the tissue clusters in Figure 8 are separable by either coordinate but with different shape: the WM and CC clusters extend further to high $\hat{\alpha}_{\text{persistence}}$ than would be predicted by a linear MK calibration, suggesting that the persistence-tail biomarker carries non-redundant information about microstructural restriction.

6 Discussion

Strengths. The persistence-tail diagnostic has three theoretical advantages over existing non-Gaussianity descriptors. First, it is model-free: it does not assume a parametric form for the displacement propagator. Second, its asymptotic behaviour is governed by a rigorous limit theorem (the parent paper’s Theorem 1) tying the exponent of the persistence lifetime tail to the stability index of the underlying α -stable propagator. Third, the computation is $O(n_{\text{vox}} n_{\text{shells}} n_{\text{dirs}} \log n_{\text{dirs}})$ and parallelises trivially over voxels.

Limitations. The simulation calibration of Figures 4–5 shows that, in the HCP-like SNR/shell-budget regime, the cumulative-sum construction (8) is partially dominated by Rician measurement noise rather than by the underlying displacement tail alone. The Hill estimator (6) therefore exhibits a known upward saturation toward $\alpha = 2$ on weakly heavy-tailed voxels, an artefact also visible in the parent paper’s Gaussian boundary [17]. The dynamic range observed in vivo ($\hat{\alpha}_{\text{persistence}} \in [1.70, 2.40]$, see Table 1) is therefore compressed relative to the full $(0, 2]$ range admissible for the underlying stability index. Two strategies plausibly extend the contrast: combining the persistence-tail map with random-matrix DW-MRI denoising [15], and using longer b -shells ($b_{\text{max}} \geq 6000 \text{ s/mm}^2$, as in the MGH Adult Diffusion protocol [10]), where

the angular attenuation footprint is more clearly separated from the noise floor. The tissue segmentation used here is intentionally simple (FA/ b_0 thresholds rather than FreeSurfer parcellation) so that the pipeline is self-contained; combining $\hat{\alpha}_{\text{persistence}}$ with a full atlas-based ROI definition is a natural extension.

Relationship to existing topology-based DW-MRI work. Previous topological-data-analysis applications to DW-MRI have focused on the persistence of *fibre-orientation distribution* fields [3, 16] or on connectomic graphs [11]. Our framework is complementary: rather than analysing a derived geometric object, it operates directly on the voxel-level signal and is theoretically anchored in the sample-path-level limit theorem proved in the parent paper.

Outlook. A natural next step is to combine $\hat{\alpha}_{\text{persistence}}$ with denoised, high- b -shell acquisitions and to evaluate clinical contrast in stroke (where ischaemic restriction sharply alters the displacement-tail structure) and tumour grading (where heterogeneous microenvironments produce mixed α -stable signatures). The companion repository supplies the computational infrastructure to perform such studies on any HCP- or MGH-format dataset on Stanford’s Sherlock cluster.

Data and Code Availability

Code. The full source code, simulation pipeline, and SLURM submission scripts are released under the MIT licence at <https://github.com/csw-research/topological-dwmri>. The parent paper’s repository (<https://github.com/csw-research/topological-levy-phase-transition>) is imported as a dependency without modification.

Data. The simulation study uses no human-subjects data; all results in Section 4 are reproducible from the code alone. The in-vivo analysis (Section 5) uses $N = 30$ healthy young-adult subjects from the Human Connectome Project S1200 release [14], available from db.humanconnectome.org under the WU-Minn HCP open-access data use terms. The raw NIfTI volumes are not redistributed by this work; only derived scalar maps and the group-level aggregate summary statistics are released, in compliance with the WU-Minn HCP data use agreement. No personally identifiable information is processed.

Reproducibility. All simulations are seeded; with the seeds and configurations recorded in `src/simulation/experiments` every figure can be regenerated by executing `python scripts/run_all_experiments.py` followed by `python analysis/make_figures.py`.

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